Module 1 Final Review

Provide the best answer for each of the following questions.

Time Remaining 01:14:11

Main Question Set

Solve six of the first ten problems to get a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though. Spend the full time and attempt every single of them!

1

Review Later

The cost of a house falls by 0.5%, ending up as $\$995,000$. How many dollars did the house cost before its price fall?

Your answer should be the number only, without commas or units.

2

Review Later

Simplify

$$1+\frac{1}{2+\frac{1}{3+\frac{1}{4}}}$$

Your answer should be a fraction without spaces. For example, if the answer were 3/2, you would enter 3 in the first blank, and 2 in the second blank.

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3

Review Later

At a potluck, Agatha brings four dishes, Bertha brings three dishes, and five other friends bring no dishes but instead money to help pay for the food. If all the dishes are eaten up, and everyone eats the same amount, what fraction of the money should go to Bertha?

Your answer should be a fraction without spaces. For example, if the answer were 3/2, you would write 3 in the first blank, and 2 in the second blank.

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4

Review Later

During your 1-hour exercise class, you spent 12 times as many minutes jogging as doing jumping jacks, and 6 times as many minutes jogging as stretching. How many minutes did you spend doing jumping jacks?

Your answer should be the number only, without units.

5

Review Later

What is the simplified value of

$$0.625 \times 4.5 \times 3.\overline{6} \times 3.2?$$

6

Review Later

Moe R. can finish cutting a lawn in 3.5 hours. Fynn Isher can cut the same lawn in 84 minutes. How many hours will it take them to mow the lawn if they work together? Your answer should be the number only, without units.

7

Review Later

Two Olympic runners are running clockwise around an indoor track. They both can run a lap in less than one minute. If they start at the same time and at the same position, after $4\frac{2}{3}$ minutes, the faster runner will have covered 1 more lap than the slower runner, with each runner having covered an integer number of laps in that time. The time it takes each of them to finish one lap is an integer number of seconds.

How many more seconds does it take the slower runner to finish one lap compared with the faster runner?

8

Review Later

Sam's little sister is playing a game called Counting Apples. Her rating for the game changes like this:

Rises by 10% from the beginning of Day 1 to the end of Day 1

Falls by 20% from the beginning of Day 2 to the end of Day 2

Rises by 30% from the beginning of Day 3 to the end of Day 3

Falls by 40% from the beginning of Day 4 to the end of Day 4

Rises by 50% from the beginning of Day 5 to the end of Day 5.

Overall, what is the percent change in her rating from the beginning of Day 1 to the end of Day 5?

Express your answer as an integer, rounded to nearest whole number.

9

Review Later

Find the value of a positive number x that makes the following true:

$$(x-10)\% \text{ of } x \text{ equals } 6.$$

10

Review Later

A red car going 40 mph catches up to a blue car going 30 mph. The blue car begins accelerating at $\frac{1}{3}\ \frac{\text{mph}}{\text{sec}}$ and the red car begins accelerating at $\frac{1}{6}\ \frac{\text{mph}}{\text{sec}}$. After two minutes, which car is ahead?

- The red car is ahead.
- The blue car is ahead.
- They are at the same position.

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

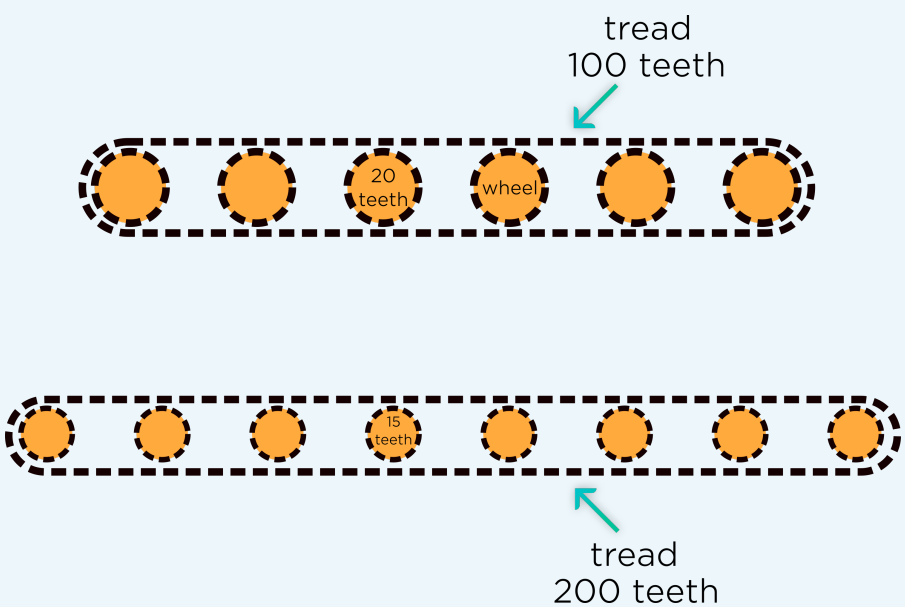
Jeremy has a pile of identical socks, each of which is either completely clean or completely dirty, and 80% of the socks in the pile are dirty. If he adds 70 more identical dirty socks to the pile, then 90% of the socks in the pile will be dirty.

How many clean socks were in the pile in the beginning?

12

Review Later

A conveyor belt 100 teeth on it, propeled by wheels that have 20 teeth. Another conveyor belt has 200 teeth, and the wheels that propel its tread have 15 teeth. Each belt has one bucket on it which is dumped once every time the belt makes one rotation, and the wheels of both rotate at the same speed. If the first bucket loader can dump $32\ \frac{\text{gallons}}{\text{hour}}$, how many gallons per hour does the second bucket loader dump?



Your answer should be the number only, without units.

13

Review Later

If 18 horses eat 5 stacks of hay in 35 days, how many horses can eat 3 stacks of hay in 9 days?

14

Review Later

The number $0.5487 \times 0.5492 \times 0.5502$ is close to $\frac{1}{n}$, where n is an integer. What is the value of n that is closest to this number?

15

Review Later

At the soccer team pizza party, the kids who came early evenly split the cost of the pizza, which altogether cost $\$120$. Four late kids came, and they pitched in to share the cost of the pizza, with each of them paying $\$0.25$ to each of the kids who were originally there. In the end, everyone paid the same amount. How many kids were there in the beginning?

16

Review Later

What is the largest fraction which is less than (and not equal to) $\frac{1}{2}$ but whose numerator and denominator are positive integers less than or equal to 100?

Your answer should be a fraction in simplest terms with no spaces, e.g. 2/3

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17

Review Later

The density of mercury at 50° degrees Celsius is $13472\ \frac{\text{kg}}{\text{m}^3}$. At 20° degrees Celsius, its density is $13545\ \frac{\text{kg}}{\text{m}^3}$. How many kilograms per meter cubed $\left(\frac{\text{kg}}{\text{m}^3}\right)$ is the density of mercury at 110° degrees Celsius?

Your answer should be the number only, without units.

18

Review Later

Sandy spent $\frac{1}{3}$ of her class period doing $\frac{4}{7}$ of her project. If she works at a constant speed, what fraction of the class time will be left after she finishes her project?

Your answer should be a fraction in simplest terms with no spaces, e.g. 2/3

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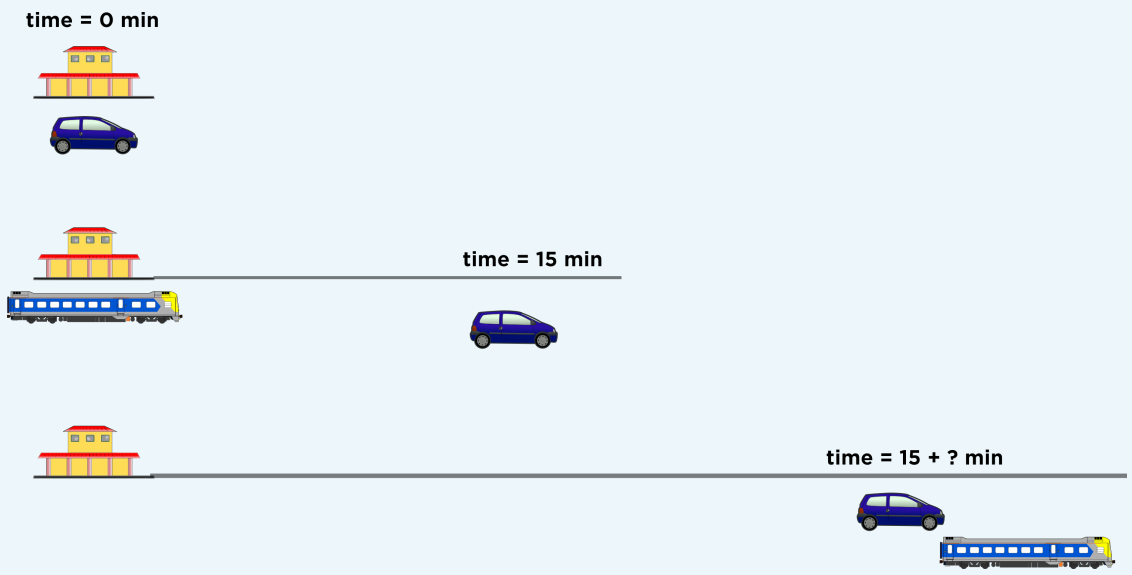
19

Review Later

A car is driving at 40 mph when it passes a train station. Fifteen minutes later, a train passes through the station at 60 mph.



How many minutes will it take for the car to be passed by the train, if both vehicles stay at a constant speed the whole time?

Your answer should be the number only, without units.

20

Review Later

You drive up a mountain at 13 miles per hour, turn around at the top, and drive back down at 39 miles per hour. How many miles per hour is your average speed for the entire trip?

Your answer should be a decimal rounded to the tenth, like 3.6, without units.

View Marked Questions

Finished

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